Does Luck Even Out? Experimental vs Theoretical Probability

SimuPhysics Probability Simulator · SimuPhysics

Year 10	~30 minutes	Mathematics · Statistics & Probability

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Simulation: https://simuphysics.com/view/mathematics-probability-simulator.html
Curriculum links: AC9M10P02 · probability · statistics

Learning intentions

- I can distinguish between experimental probability and theoretical probability.
- I can design and run a repeated chance simulation and record the results.
- I can explain why experimental results get closer to the theoretical probability as the number of trials increases.

Getting started

Open the SimuPhysics Probability Simulator and choose a simple experiment (e.g. flipping a coin or rolling a die). Set the trial count to a small number first, before you read any further.

Tasks

1. Run 10 trials of a coin flip. Record the number of heads and tails, and the experimental probability of heads (as a fraction or percentage).

2. Compare your result to the theoretical probability of getting heads (1/2, or 50%). Was your experimental result close to this, or quite different?

3. Run the same coin-flip experiment at three different trial counts (e.g. 10, 100, 1000). Record the experimental probability of heads each time.

Number of trials	Experimental probability of heads

4. Describe the pattern in your table. What happens to the experimental probability as the number of trials increases?

5. Switch to a six-sided die. Run 60 trials and record the experimental probability of rolling a 6.

6. Explain, in your own words, the difference between 'experimental probability' and 'theoretical probability', using your die results as an example.

7. A classmate rolls a die 6 times and gets zero 6s. They say 'this die must be broken.' Using what you've learned from the simulation, explain why they might be wrong to conclude this from only 6 trials.

Extension (optional)

A casino game relies on the same idea — small numbers of trials can look 'unfair', but large numbers of trials behave predictably. Use your simulation results to explain why this matters for how a casino makes reliable profits over time.